

Pyramidal Multiple Instance Detection Network With Mask Guided Self-Correction for Weakly Supervised Object Detection

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Abstract—Weakly supervised object detection has attracted more and more attention as it only needs image-level annotations for training object detectors. A popular solution to this task is to train a multiple instance detection network (MIDN) which integrates multiple instance learning into a deep convolutional neural network. One major issue of the MIDN is that it is prone to be stuck at local discriminative regions. To address this local optimum issue, we propose a pyramidal MIDN (P-MIDN) comprised of a sequence of multiple MIDNs. In particular, one MIDN performs proposal removal for its subsequent MIDN to reduce the exposure of local discriminative proposal regions to the latter during training. In this manner, it allows our MIDNs to focus on proposals which cover objects more completely. Furthermore, we integrate the P-MIDN into an online instance classifier refinement (OICR) framework. Combined with the P-MIDN, a mask guided self-correction (MGSC) method is proposed to generate high-quality pseudo ground-truths for training the OICR. Experimental results on PASCAL VOC 2007, PASCAL VOC 2010, PASCAL VOC 2012, ILSVRC 2013 DET and MS-COCO benchmarks demonstrate that our approach achieves state-of-the-art performance.

Index Terms—Weakly supervised learning, object detection, pyramidal network.

I. INTRODUCTION

WITH recent development of deep convolutional neural networks (CNNs), significant progress has been made in object detection [1]–[5]. However, these detectors require a large number of images with bounding-box annotations for model training. It is labor-intensive and time-consuming to collect bounding-box annotations. Because of the difficulty of collecting these training data, weakly supervised object

detection (WSOD), which requires only image-level annotations to train object detectors, has attracted more and more attention in recent years [6]–[11]. Compared with bounding-box annotations, image-level annotations are much easier to be collected. For example, there is an enormous number of images with class tags available on the internet.

Most previous works [6]–[15] formulate weakly supervised object detection as a multiple instance learning (MIL) problem. In [6], a multiple instance detection network (MIDN) is proposed to integrate the MIL formulation into a deep CNN. The MIDN is learned in an end-to-end manner with image-level annotations for scoring object proposals. The MIDN suffers a local optimum issue since it is prone to be stuck at local discriminative regions (e.g. heads of animals), which is not desirable for object detection, as illustrated in Fig. 1(a). To address this issue, many works have been proposed by exploiting segmentation [8], [10], [16]–[18], context [19], [20] or other information [21]–[23]. A recent work, coupled MIDN [10], learns two MIDNs in a sequential order such that the proposals around the top-scoring region (usually a local discriminative region) chosen by the first MIDN are removed before training the second MIDN. This reduces the exposure of local discriminative proposals to the second MIDN during training. The integration of the coupled MIDN into an online instance classifier refinement (OICR) framework [7] achieves promising performance for weakly supervised object detection. However, when there are multiple objects of the same class in an image, the coupled MIDN may fail as it can only remove local discriminative proposals from one object (see Fig. 1(b)). The second MIDN might be stuck at a local discriminative region of the other object.

Inspired by the coupled MIDN [10], we propose a pyramidal MIDN to address the local optimum issue of the MIDN. The pyramidal MIDN consists of several MIDNs which are trained sequentially. One MIDN performs proposal removal for its subsequent MIDN as in the Coupled MIDN. Different from the coupled MIDN that removes proposals around the top-scoring regions, we suppress proposals according to bounding-box intersection over union (IOU). As ground-truth bounding-box annotations are not available, we obtain coarse object bounding boxes from segmentation masks generated by a weakly supervised segmentation method (e.g. [24]). In the beginning, the first MIDN selects a top-scoring proposal. Then, it discards proposals with IOUs smaller than a threshold.

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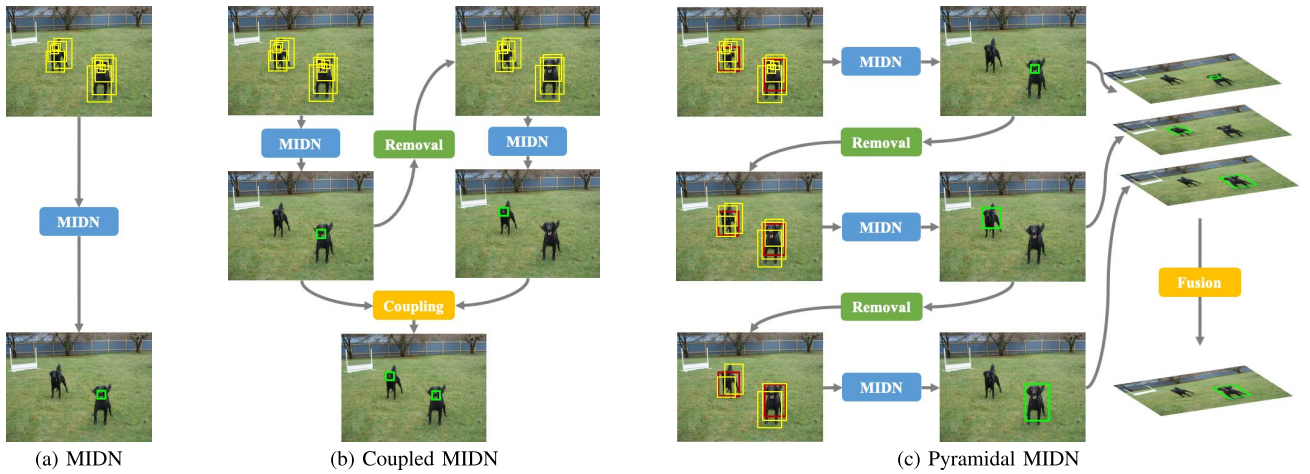


Fig. 1. Comparison of different multiple instance detection networks. Yellow bounding boxes denote object proposals. Green bounding boxes denote the top-scoring proposals selected as pseudo ground-truths. Red bounding boxes are obtained from segmentation masks, and are used to compute IOUs of proposals. Only parts of object proposals are displayed for clarity.

The threshold is chosen adaptively based on the IOU between the top-scoring proposal and the bounding box derived from the segmentation mask. Subsequently, the next MIDN will be trained with proposals of higher IOUs. The subsequent MIDNs perform proposal removal in a similar way. Note that, our proposal removal approach discards local object parts and overly large proposals in a pyramidal manner (*i.e.* in increasing order of IOU). As illustrated in Fig. 1(c), it can well address the situation when multiple objects of the same class appear in an image.

We integrate our pyramidal MIDN, dubbed P-MIDN, into the popular OICR framework [7] for weakly supervised object detection. The conventional OICR framework consists of one MIDN and an online refinement module with several instance classifiers. In the OICR, the MIDN provides initial pseudo bounding-box ground-truths (GTs) for the first instance classifier. As reported in [8], [10], [20], the initial pseudo GTs for the online refinement module are critical for the OICR to achieve good performance. To generate high-quality initial pseudo GTs, we replace the MIDN in the OICR with our P-MIDN and propose a mask guided self-correction (MGSC) method. The MGSC is designed to correct poor pseudo GTs provided by the P-MIDN via our proposed self-labeling and local refinement mechanisms. We demonstrate the effectiveness of our proposed approach on PASCAL VOC 2007, PASCAL VOC 2010, PASCAL VOC 2012, ILSVRC 2013 DET and MS-COCO benchmarks. Experimental results show that our method outperforms the state-of-the-art on all five datasets.

Overall, the main contributions of the paper are three-fold:

- We propose a pyramidal MIDN to alleviate the local optimum issue of the conventional MIDN effectively. Compared with the Coupled MIDN, our pyramidal MIDN can better tackle multiple instances of the same class in an image.
- We develop a mask guided self-correction (MGSC) method to correct the initial pseudo GTs from P-MIDN via our proposed self-labeling and local refinement mechanisms. Using MGSC, we obtain high-quality

pseudo GTs for training the first instance classifier in the online refinement module.

- Our proposed approach achieves the state-of-the-art performance on five commonly used benchmarks for weakly supervised object detection.

II. RELATED WORK

The majority of existing works formulate weakly supervised object detection (WSOD) as a multiple instance learning (MIL) [25] problem. One major issue of the MIL paradigm is that it is prone to be trapped at local optima. Some works [13], [26], [27] attempt to develop better initialization methods, while other works [9], [12], [28], [29] cope with the local optima from the viewpoint of optimization. Prior knowledge is used to constrain the MIL optimization, such as pairwise similarity [11], [14], contextual information [19], [20], segmentation [8], [10], [18], low-level objectness [30], motion [22], [31], *etc.* Moreover, some recent works also exploit other forms of weak supervision, such as per-class object count [21], image caption [32], scribbles [33], human-object interaction [23], [34], *etc.* In this paper, we focus on the WSOD task with only image-level annotations.

Recently, state-of-the-art WSOD methods have achieved appealing performance by incorporating the MIL and CNN into a unified framework. Bilen and Vedaldi [6] transform the WSOD task into a multi-label image classification problem and propose a multiple instance detection network (MIDN) with two parallel branches of classification and detection. Tang *et al.* [7] propose an online instance classifier refinement (OICR) approach by combining MIDN and multi-stage instance classifiers, which partly alleviates the local optimum issue. In [9], [28], CNNs are learned on groups of spatially overlapping and class-related proposals for both object discovery and localization with image-level annotations. In order to localize bounding boxes around objects, Shen *et al.* [35] re-weight proposals by a Category-aware Spatial Constraint (CSC) scheme. Kosugi *et al.* [20] train an auxiliary context classifier to select proposals of high overlap with object instances as pseudo ground-truths (GTs) for the

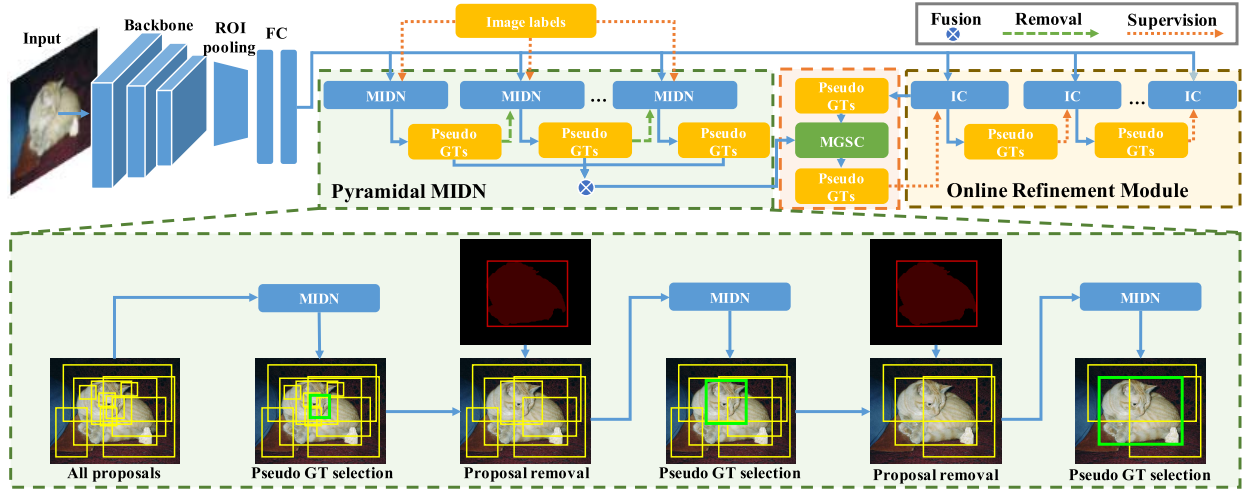


Fig. 2. Overview of our proposed method. The top part is the overall structure of our proposed method. The bottom part illustrates the Pyramidal MIDN. The red bounding box is the smallest rectangle enclosing the segmentation mask. Green bounding boxes indicate the selected pseudo ground truths. Yellow bounding boxes are object proposals.

OICR approach. From a different perspective, Zhang *et al.* [36] investigate how to distill better pseudo GTs from OICR [7] to train a fully supervised detector. Yang *et al.* [37] and Zeng *et al.* [30] jointly optimize a bounding-box regressor and an object detector in an end-to-end manner with image-level annotations. Following the common practice of most state-of-the-art approaches, we adopt the OICR [7] as our framework. Cheng *et al.* [38] propose a high-quality proposal generation technique by combining selective search [39] and Gradient-weighted Class Activation Mapping (Grad-CAM) [40].

Weakly supervised segmentation masks have been utilized to improve the performance of the WSOD task. Diba *et al.* [16] and Wei *et al.* [8] introduce segmentation confidence maps as guidance for better instance proposal selection. The methods [41], [42] jointly learn detection and segmentation with image-level annotations in a multi-task learning framework. Cyclic collaboration of detection and segmentation is explored in [17], [18] to exploit the complementarity between weakly supervised detection and segmentation approaches: a weakly supervised detection approach often produces partial detection results, while segmentation masks from a weakly supervised segmentation approach can usually better cover entire objects but tend to include background noises. Gao *et al.* [10] propose a coupled MIDN, where local discriminative regions selected by the first MIDN are removed with the guidance of segmentation masks before training the second MIDN. However, the coupled MIDN tends to fail when there are multiple objects of the same class in an image. Inspired by the coupled MIDN, we exploit segmentation masks to construct a pyramid of MIDNs to better address the local optimum issue, especially when multiple objects of the same class appear in an image. Moreover, we further propose a mask guided self-correction method to generate better pseudo GTs for iterative refinement.

III. PROPOSED METHOD

Our proposed method is based on a widely used weakly supervised object detection framework, namely online instance

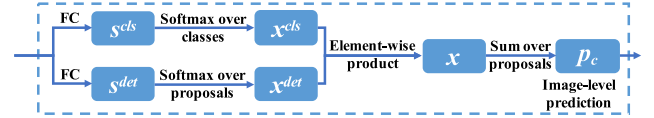


Fig. 3. Illustration of a basic MIDN.

classifier refinement (OICR) [7]. The overview of our proposed network is illustrated in Fig. 2. The pyramidal multiple instance detection network (P-MIDN) consists of a pyramid of several MIDNs followed by an online refinement module. At the training stage, our MIDNs are trained with fewer and fewer proposals in a pyramidal manner and used to generate pseudo ground-truths for the subsequent online refinement module. An instance classifiers (IC) in the online refinement module is trained by the pseudo ground-truths provided by its previous one or P-MIDN.

A. Preliminaries

The work [6] proposes an MIDN for weakly supervised object detection. The MIDN combines multiple instance learning (MIL) with a CNN in an end-to-end framework as shown in Fig. 3. For an input image I , let $R = \{r_1, r_2, \dots, r_N\}$ be a set of object proposals generated by a region proposal method (e.g. selective search [39]), and $Y = \{y_1, y_2, \dots, y_C\} \in \mathbb{R}^C$ be the image-level annotations, where $y_c \in \{0, 1\}$ indicates the presence or absence of the object class c in an image. The image I and proposals R are first fed to the network backbone followed by an ROI pooling layer to extract feature vectors for the proposals. Then, the feature vectors are passed through two parallel branches to generate two score matrices $s^{cls} \in \mathbb{R}^{C \times N}$ and $s^{det} \in \mathbb{R}^{C \times N}$. The score matrices are normalized by Softmax along the category and proposal directions respectively: $x^{cls} = \frac{\exp(s_{c,n}^{cls})}{\sum_c \exp(s_{c,n}^{cls})}$ and $x^{det} = \frac{\exp(s_{c,n}^{det})}{\sum_n \exp(s_{c,n}^{det})}$. The scores of the proposals are computed by an element-wise product of the two score matrices: $x = x^{cls} \odot x^{det}$. Finally, the probability of the

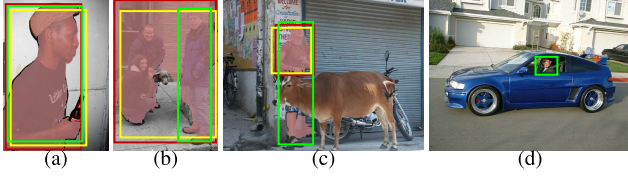


Fig. 4. Examples of IOU fitness. Red bounding boxes indicate the smallest rectangles enclosing the segmentation masks. Yellow bounding boxes indicate object proposals with high IOU fitness. Green bounding boxes are high-quality pseudo GTs. Only the masks of the person class are displayed.

image containing the c -th object class is defined by the sum of proposal scores: $p_c = \sum_{n=1}^N x_{c,n}$. The MIDN is trained by minimizing the binary cross-entropy loss:

$$L_{midn} = - \sum_{c=1}^C (y_c \log p_c + (1 - y_c) \log(1 - p_c)). \quad (1)$$

At the inference stage, the proposal scores are taken as detection confidences.

B. Pyramidal MIDN

Since an MIDN is trained with only image-level supervision and no additional constraints are imposed, it tends to detect local discriminative regions (e.g. heads of people and animals) rather than whole objects, as pointed out in [7], [43]. It has been manifested in [8], [17], [18], [42] that weakly supervised segmentation can better discover complete objects, which could be supplementary to the MIDN for escaping from local discriminative regions. In a recent work, dubbed coupled MIDN (C-MIDN) [10], proposals around the highest-scoring proposal (usually a local discriminative region) selected by the first MIDN are carefully removed before training the second one in order to alleviate the issue of being stuck at local discriminative regions. However, this method is prone to fail when there are multiple objects of the same class in an image. This is because local discriminative proposals from other objects still remain for training the second MIDN as illustrated in Fig. 1(b). Motivated by those works, we propose a pyramidal MIDN (P-MIDN) utilizing both segmentation and proposal removal to learn multiple MIDNs that find ground-truth object regions in different degrees (i.e. scales).

Our proposed P-MIDN consists of K sequential MIDNs as shown in the top part of Fig. 2. The first MIDN is trained with all proposals as described in Sec. III-A. For the k -th MIDN ($1 < k \leq K$), some proposals are removed according to the highest-scoring proposals (which we call pseudo ground-truths (GTs)) from the $(k-1)$ -th MIDN. For the input image I , let M_c be the segmentation mask for object class c obtained by a weakly supervised segmentation method [24]. We divide the segmentation mask M_c into connected components. We define the fitness of a proposal to a connected component by the IOU between the proposal and the tightest bounding box around the connected component. Fig. 4 shows different cases of high IOU fitness.

Denote by $f_{n,c}$ the highest IOU between the proposal r_n and all connected components in M_c . $f_{n,c}$ reflects the quality

of the proposal r_n as a potential pseudo GT for object class c if its corresponding connected component well cover an object of class c (see Fig. 4(a)). Let $\tilde{r}_c^{k-1}$ be the pseudo GT selected by the $(k-1)$ -th MIDN for object class c . When performing proposal removal for the k -th MIDN, we discard the proposals with $f_{n,c} < \min(\alpha \tilde{f}_c^{k-1}, \tau)$, where $\alpha \geq 1$ is a scaling factor, $\tilde{f}_c^{k-1}$ is the highest fitness of the pseudo GT $\tilde{r}_c^{k-1}$ and τ is a fitness upper bound. In some cases, there are several objects of the same class which are adjacent to each other in an image. These objects probably fall in a single connected component (see Fig. 4(b)). For such a connected component, a high-quality proposal (e.g. the green bounding box in Fig. 4(b)) may not correspond to a very high fitness. To retain such proposals for training k -th and its subsequent MIDNs, we do not remove a proposal, when its IOU fitness is higher than τ . A similar case could happen when an object is separated into several connected components as shown in Fig. 4(c). The upper bound τ serves to balance between single-object, multi-object and broken connected components. According to our proposal removal procedure, the K MIDNs in the P-MIDN are trained with proposals of increasing IOU fitness. The P-MIDN is trained by minimizing the following objective:

$$\begin{aligned} L_{p-midn} &= - \sum_{k=1}^K L_{midn}^k \\ &= - \sum_{k=1}^K \sum_{c=1}^C (y_c \log p_c^k + (1 - y_c) \log(1 - p_c^k)), \end{aligned} \quad (2)$$

where p_c^k is the probability predicted by the k -th MIDN as defined in Sec. III-A.

Compared with the C-MIDN, our proposed P-MIDN can better tackle images containing multiple objects of the same class as illustrated in Fig. 1(c). The C-MIDN only removes proposals around a pseudo GT as shown in Fig. 1(b). In contrast, the P-MIDN removes proposals with IOU fitness below an adaptively chosen threshold at each stage. In this way, proposals with similar fitness are removed. For instance, in Fig. 1(c), the proposals around the head regions of the two dogs have similar fitness and thus are removed simultaneously.

C. Pseudo GT Generation With Mask Guided Self-Correction

The P-MIDN provides initial pseudo GTs for the first instance classifier (IC) in the online refinement module as shown in the top-right part of Fig. 2. Here, we combine the proposal score and IOU fitness provided by P-MIDN and segmentation masks to generate pseudo GTs. To be specific, given an object class c in an image, each of the K MIDNs selects one pseudo GT with the maximum value of $x_{n,c}^k f_{n,c}$. In general, local discriminative regions have a high proposal score $x_{n,c}^k$ but a low IOU fitness $f_{n,c}$. The IOU fitness serves to penalize the local discriminative regions, thus preventing them from being selected as pseudo GTs. Then, the best one is chosen from the K pseudo GTs as a pseudo GT for the first instance classifier.

As the masks are generated with only image-level supervision, they are usually not perfect (e.g. covering multiple objects, missing object parts or losing objects as shown

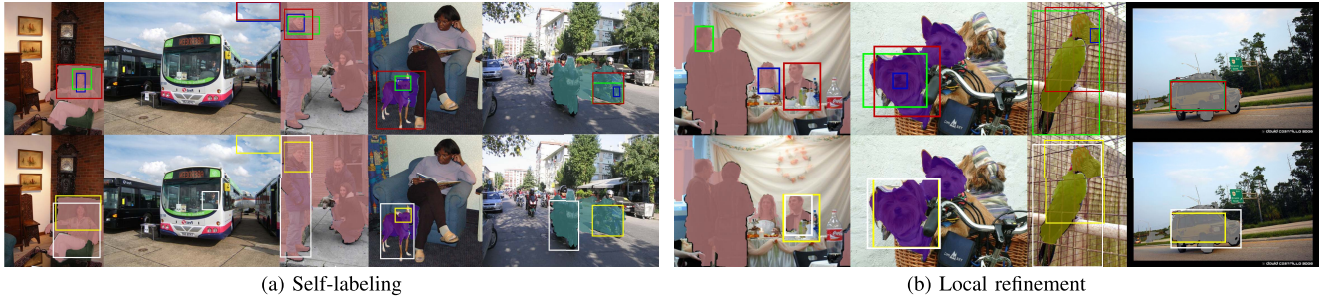


Fig. 5. Mask guided self-correction. The first row shows the top-scoring proposals selected by the three MIDNs in the P-MIDN (blue, green and red bounding boxes respectively). The second row shows initial pseudo GTs before correction (yellow bounding boxes) and after correction (white bounding boxes).

in Fig. 4(b-d)). This affects both the training of the P-MIDN and the selection of the initial pseudo GTs. The top row of Fig. 5 shows some top-scoring proposals selected by the P-MIDN. However, the selected initial pseudo GTs are unsatisfactory as indicated by yellow bounding boxes in the bottom row of Fig. 5. Driven by this issue, we propose a mask guided self-correction (MGSC) method for the first IC in the online refinement module. MGSC performs self-labeling and local refinement to modify initial pseudo GTs from the P-MIDN.

1) *Self-Labeling*: We take the IOU fitness as the quality indicator for pseudo GTs. If the selected pseudo GT from the P-MIDN has an IOU fitness lower than the fitness upper bound τ defined above, we consider it as a poor pseudo GT and discard it. Instead, we enable the first instance classifier to select the pseudo GT for itself. We call this operation *self-labeling*. Self-labeling is particularly helpful when some local discriminative regions still predominate (third and fourth columns of Fig. 5(a)). Moreover, self-labeling can correct the noisy pseudo GTs when the segmentation mask is absent. For instance, the segmentation of the driver is missing in the second column of Fig 5(a). We adopt a warm-up stage to train the first instance classifier with pseudo GTs of high IOU fitness ($f_{n,c} \geq \tau$) as described in Sec. III-E. Thus, it is trained to achieve the capability of selecting high-fitness pseudo GTs.

2) *Local Refinement*: Among high-fitness proposals ($f_{n,c} \geq \tau$), some may not well cover entire objects due to imperfect segmentation (See Fig. 4(b-c)). To localize better pseudo GTs, we again resort to the first instance classifier. Specifically, we employ the first instance classifier to score the proposals that have an IOU higher than the IOU threshold β (e.g. $\beta = 0.7$) with the pseudo GT from the P-MIDN. Among these proposals, the highest-scoring proposal is selected. We call this operation *local refinement*. With the proposed P-MIDN, a large portion of high-quality pseudo GTs are selected for training the first instance classifier. Therefore, the instance classifier is expected to be capable of distinguishing high-quality proposals from low-quality ones. Note that the local refinement operation is only applied to high-fitness candidate pseudo GTs from the P-MIDN. Fig. 5(b) shows some examples of local refinement. After local refinement, better pseudo GTs are obtained.

D. Online Instance Classifier Refinement

The online refinement module in [7] performs progressive instance classifier refinement. Assume that there are $\hat{K}$

branches in this module. Each branch is an instance classifier (IC). The first IC takes the pseudo GTs from the P-MIDN as training examples. For $\hat{k} > 1$, the $\hat{k}$ -th IC takes the pseudo GTs from the $(\hat{k} - 1)$ -th IC as training examples. For a set of proposals, the $\hat{k}$ -th IC outputs a probability matrix $\hat{x}^{\hat{k}} \in \mathbb{R}^{(C+1) \times N}$ with the extra $(C + 1)$ -th class representing background and $\hat{x}_{n,c}^{\hat{k}}$ is the score of the proposal r_n for object class c . Given an image with the presence of object class c , the highest scoring proposal $r_{n_c}^{\hat{k}}$ selected by the $(\hat{k} - 1)$ -th IC is taken as the pseudo GT for the k -th IC, where $n_c^{\hat{k}} = \arg \max_n \hat{x}_{c,n}^{\hat{k}-1}$. With pseudo GTs, each instance classifier is trained as in Fast R-CNN [44] with two differences that (1) no bounding-box regression is involved and (2) each proposal is assigned a loss weight. The cross-entropy loss for the $\hat{k}$ -th IC is defined by:

$$L_{refine}^{\hat{k}} = -\frac{1}{|N|} \sum_{n=1}^N \sum_{c=1}^{C+1} w_{c,n}^{\hat{k}} y_{c,n}^{\hat{k}} \log(x_{c,n}^{\hat{k}}), \quad (3)$$

where $w_{c,n}^{\hat{k}} = \hat{x}_{c,n_c^{\hat{k}-1}}^{\hat{k}-1}$ is the loss weight and $y_{c,n}^{\hat{k}} \in \{0, 1\}$ is the label for the proposal r_n determined by the IOU with the pseudo GT $r_{n_c^{\hat{k}}}$. The overall training loss for the online refinement module is defined by:

$$L_{oicr} = \sum_{\hat{k}=1}^{\hat{K}} L_{refine}^{\hat{k}}. \quad (4)$$

We refer readers to [7] for more details on the online refinement module.

E. Network Optimization

The overall network is jointly optimized by minimizing the following training loss:

$$L_{total} = \lambda L_{p-midn} + L_{oicr}, \quad (5)$$

where λ is a parameter for balancing L_{p-midn} and L_{oicr} . The proposed MGSC method for pseudo GT generation requires the first instance classifier to select high quality pseudo GTs. We introduce a warm-up stage to train the first instance classifier only with pseudo GTs of high IOU fitness from the P-MIDN. Specifically, at the beginning, we train the P-MIDN and the first instance classifier for T iterations with the following loss:

$$L_{warmup} = \lambda L_{p-midn} + L_{refine}^1. \quad (6)$$

IV. EXPERIMENTS

A. Datasets and Metrics

We evaluate our proposed approach on five widely used object detection benchmarks, *i.e.*, PASCAL VOC 2007 [45], PASCAL VOC 2010, PASCAL VOC 2012 [46], ILSVRC 2013 DET [47] and MS-COCO [48].

The PASCAL VOC benchmarks are all annotated with 20 object classes. For the PASCAL VOC benchmarks, the *trainval* set (about 5K images for 2007, 10K images for 2010, and 11K images for 2012) with only image-level annotations are used for training, and the *test* set is used for evaluation. For PASCAL VOC benchmarks, mean average precision (mAP) on the *test* set and correct localization (CorLoc) on the *trainval* set are adopted for evaluation. Following the standard PASCAL VOC criterion [47], a predicted bounding box is considered to be true detection if it has an $\text{IOU} \geq 0.5$ with a ground-truth annotation.

The ILSVRC 2013 DET dataset is a large scale dataset with 200 categories. Following [49], we split the *val* set into *val1* and *val2*, and randomly sample at most 1K images per class from *train* set (dubbed *train1k*). We use *val1*+*train1k* as well as *val1* (about 140K and 10K images) for training. We test the proposed method on *val2* split using the mAP metric.

The MS-COCO benchmark contains 80 categories. Our models are trained on the *train2014* split (about 83K images) with only image-level annotations, and evaluated on *val2014*, *minival* and *test-dev* splits. For MS-COCO, AP_{50} and AP are reported, which denote average precision at IOU threshold of 0.5 and mean average precision over the IOU thresholds [0.5:0.05:0.95] respectively.

B. Implementation Details

We implement and test our method using PaddlePaddle and PyTorch. The VGG16 [50] pre-trained on ImageNet [47] is adopted as our network backbone. We use $K = 3$ MIDNs and $\hat{K} = 3$ instance classifiers in our network. The training batch size is set to 4. We train our network for 100K, 200K, 200K, 300K and 250K iterations on PASCAL VOC 2007, PASCAL VOC 2010, PASCAL VOC 2012, ILSVRC 2013 DET and MS-COCO respectively. We adopt a warm-up stage T of 40K, 80K, 80K, 120K and 100K respectively to train the first instance classifier on the five benchmarks with the loss function in Eq. (6). Stochastic gradient descent is adopted for network optimization with momentum 0.9, initial learning rate 0.001 and weight decay 0.0001. The learning rate is reduced by a factor of 10 after 60K, 120K, 120K, 180K and 150K.

Similar to [10], [43], we also adopt selective search [39] for the PASCAL VOC benchmarks and ILSVRC 2013 DET *train1k*, and MCG [51] for MS-COCO and ILSVRC 2013 DET *val* to generate about 2000 object proposals. For segmentation mask generation, we adopt a state-of-the-art weakly supervised segmentation method IRN [24], which is trained on the same training images, following the default settings. Horizontal flipping and multi-scale data augmentation techniques are adopted in both training and testing.

TABLE I

COMPARISON OF mAP AND CorLoc PERFORMANCE (%) ON PASCAL VOC 2007

Method	mAP	CorLoc
OICR-VGG16 [7]	41.2	60.6
PCL-VGG16 [43]	43.5	62.7
TS ² C [8]	44.3	61.0
WSRPN [52]	45.3	63.8
C-WSL [21]	46.8	63.5
MELM [53]	47.3	61.4
CSC [35]	43.0	62.2
OICR+W-RPN [22]	46.9	66.5
C-MIL [9]	50.5	65.0
WS-JDS [17]	45.6	64.5
CAP-SRN [20]	47.6	66.7
SDCN [18]	50.2	68.6
C-MIDN [10]	52.6	68.7
PG-PS [38]	51.1	69.2
OIM+IR [11]	50.1	67.2
P-MIDN+MGSC	53.9	69.8
OICR-Ens.+FRCNN [7]	47.0	64.3
C-SPCL [54]	45.9	62.4
PCL-Ens.+FRCNN [43]	48.8	66.6
TS ² C+FRCNN [8]	48.0	-
WSRPN-Ens.+FRCNN [52]	50.4	68.4
C-WSL+FRCNN [21]	48.2	66.1
WSD+PGE+PGA+FSD2+IGL [55]	53.1	71.2
CSC+FRCNN [35]	49.0	66.1
C-MIL+FRCNN [9]	53.1	-
PredNet [29]	52.9	70.9
WS-JDS+FRCNN [17]	52.5	68.6
CAP-SRN+PGE+FRCNN [20]	52.1	-
SDCN+FRCNN [18]	53.7	72.5
C-MIDN+FRCNN [10]	53.6	71.9
WSOD ² [30]	53.6	69.5
PCL+GAM+REG [37]	51.5	68.0
Label-PEnet [42]	53.1	68.2
PG-PS+FRCNN [38]	54.6	72.2
OIM+IR+FRCNN [11]	52.6	68.8
P-MIDN+MGSC+FRCNN	55.0	72.4

For training, short edges of input images are randomly re-scaled to a scale in {480, 576, 688, 864, 1200}, and the longest image edges are capped to 2000. For testing, input images are augmented with all five scales and horizontal flipping. For all experiments, the threshold for NMS is set to 0.3. Proposals having a IOU larger than 0.5 with a selected GT are sampled as positive examples, while negative proposals have a IOU between 0.1 and 0.5. We set the loss weight λ of L_{p-midn} in Eq. (5) to 0.2. Empirically, the scaling factor α , the fitness upper bound τ and the IOU threshold β are set to 1.3, 0.4 and 0.7 respectively.

C. Comparison With State-of-the-Arts

We compare our proposed method with recent works [8], [10], [17], [18], [42] that also leverage segmentation for weakly supervised object detection, as well as other state-of-the-arts. Following the common practice of the WSOD task, we compare our proposed method with others in two settings. In the first setting, we compare our method with other WSOD models without using bounding box regression. In the second setting, we use our model to generate pseudo GTs for training a Fast R-CNN detector [44].

1) *Results on PASCAL VOC 2007*: Tab. I shows the mAP performance of all the competing approaches on the *test* set.

TABLE II
COMPARISON OF mAP AND CorLoc PERFORMANCE (%)
ON PASCAL VOC 2010

Method	mAP	CorLoc
WCCN [16]	36.4	-
CSC [35]	37.6	61.0
MELM [53]	39.9	-
WS-JDS [17]	39.9	63.1
P-MIDN+MGSC	53.5¹	74.3
WCCN+FRCNN [16]	37.9	-
SP [56]	40.3	-
C-SPCL [54]	39.1	-
GAL-fWSD300 [57]	45.1	68.3
CS+AS+MIL+Seg+FT [58]	36.3	-
CSC+FRCNN [35]	43.2	66.2
ODGA+SSD [59]	45.7	-
WS-JDS+FRCNN [17]	45.7	68.1
P-MIDN+MGSC+FRCNN	53.8²	76.7

The top part of the table shows the performance of models trained in the first setting. In this setting, our method achieves the best performance among all the methods. Particularly, our method improves the performance over the second best method, C-MIDN, by 1.3% in mAP. The C-MIDN also exploits segmentation and proposal removal to address the local optimum issue of the OICR. This result demonstrates the effectiveness of our proposed P-MIDN and MGSC for pseudo GT generation. In the bottom part of the table, models are trained in the second setting. Our Fast R-CNN model achieves the best performance *i.e.*, 55.0% mAP among all the methods. Our method (P-MIDN+MGSC+FRCNN) outperforms two strong methods, C-MIDN+FRCNN and SDCN+FRCNN, by 1.4% and 1.3% in mAP, respectively. This validates the superiority of our proposed approach for generating pseudo GTs in training the Fast R-CNN detector. Tab. I also summarizes the CorLoc performance on `trainval` set. In the first setting, our proposed approach outperforms the C-MIDN by 1.1%. In the second setting, our Fast R-CNN attains on par performance compared to the state-of-the-art.

2) *Results on PASCAL VOC 2010*: Tab. II shows that our method achieves 53.5% mAP and 74.3% CorLoc on the PASCAL VOC 2010 and outperforms the others significantly. After training a Fast R-CNN using the pseudo GTs produced by our trained network (P-MIDN+MGSC), we achieve consistent improvements and the best performance, *i.e.*, 53.8% mAP and 76.7% CorLoc on the PASCAL VOC 2010.

3) *Results on PASCAL VOC 2012*: Experimental results on the PASCAL VOC 2012 dataset are reported in Tab. III. Our proposed method achieves an mAP of 52.8%, which exceeds the results of the state-of-the-art. As for the CorLoc metric, the performance of our approach reaches 73.3%, which is 2.1% higher than that of the most competitive C-MIDN. By training a Fast R-CNN using the pseudo GTs generated by P-MIDN+MGSC, our method achieves the highest mAP and CorLoc *i.e.*, 53.6% and 76.1% respectively.

TABLE III
COMPARISON OF mAP AND CorLoc PERFORMANCE (%)
ON PASCAL VOC 2012

Method	mAP	CorLoc
OICR-VGG16 [7]	37.9	62.1
PCL [43]	40.6	63.2
TS ² C [8]	40.0	64.4
WSRPN [52]	40.8	64.9
MELM [53]	42.4	64.9
CSC [35]	37.1	61.4
OICR+W-RPN [22]	43.2	67.5
C-MIL [9]	46.7	67.4
WS-JDS [17]	39.1	63.5
CAP-SRN [20]	43.4	66.7
SDCN [18]	43.5	67.9
C-MIDN [10]	50.2	71.2
PG-PS [38]	48.3	68.7
OIM+IR [11]	45.3	67.1
P-MIDN+MGSC	52.8³	73.3
OICR-Ens.+FRCNN [7]	42.5	65.6
PCL-Ens.+FRCNN [43]	44.2	68.0
TS ² C+FRCNN [8]	44.4	-
WSRPN-Ens.+FRCNN [52]	45.7	69.3
WSD+PGE+PGA+FSD2+IGL [55]	49.4	71.1
CSC+FRCNN [35]	44.1	67.0
PredNet [29]	48.4	69.5
WS-JDS+FRCNN [17]	46.1	69.5
CAP-SRN+PGE+FRCNN [20]	47.9	-
SDCN+FRCNN [18]	46.7	69.5
C-MIDN+FRCNN [10]	50.3	73.3
WSOD ² [30]	47.2	71.9
PCL+GAM+REG [37]	45.6	68.7
Label-PEnet [42]	49.2	71.3
PG-PS+FRCNN [38]	52.9	71.0
OIM+IR+FRCNN [11]	46.4	69.5
P-MIDN+MGSC+FRCNN	53.4⁴	76.0

4) *Results on ILSVRC 2013 DET*: Experimental results on the more challenging large-scale ILSVRC 2013 DET dataset are reported in Tab. IV. (i) When the `val1` split is used as training data, our method achieves 16.3% mAP. (ii) When the `train1k+val1` split is used, our method achieves the best performance *i.e.*, 29.0% mAP. By further training a Fast R-CNN, our method obtains the best mAP performance on both training splits and outperforms the state-of-the-art by a large margin of 12.3%.

5) *Results on MS-COCO*: To further demonstrate the effectiveness of our proposed method, we also conduct experiments on another large-scale dataset MS-COCO. As shown in Tab. V, our proposed method achieves AP₅₀ of 27.4% and AP of 13.1%, surpassing other state-of-the-art methods by a large margin on the `val2014` split. After training a Fast R-CNN using the pseudo GTs, our approach achieves the best AP₅₀ of 28.5% and AP of 13.2%. We also report our results on the `minival` and `test-dev` splits. Note that the `train2014` split is a subset of `train2017` or `trainval35k` splits. Our proposed approach trained on the `train2014` split (around 83K images) even outperforms all the other methods trained on

¹<http://host.robots.ox.ac.uk:8080/anonymouse/XEAQLL.html>

²<http://host.robots.ox.ac.uk:8080/anonymouse/MKCZKB.html>

³<http://host.robots.ox.ac.uk:8080/anonymouse/TFVDFU.html>

⁴<http://host.robots.ox.ac.uk:8080/anonymouse/DQIJGD.html>

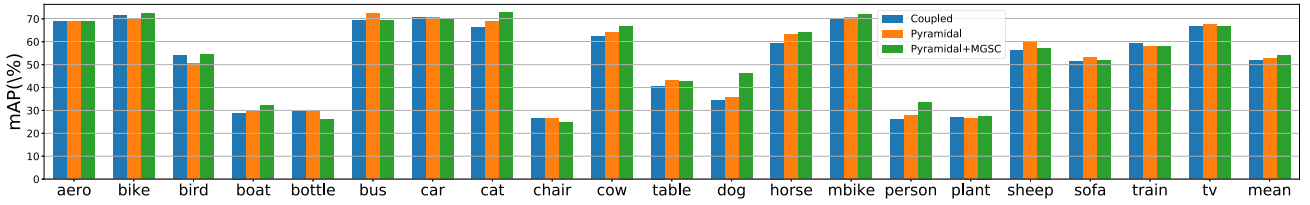


Fig. 6. Per-class detection performance of coupled MIDN and pyramidal MIDN on PASCAL VOC 2007 test set.

TABLE IV

COMPARISON OF mAP PERFORMANCE (%) ON ILSVRC 2013 DET

Method	training data	testing data	mAP
MILinear [60]	-	-	9.6
WCCN [16]	-	-	14.6
PCL [43]	train1k+val1	val2	18.4
WSRPN [52]	train1k+val1	val2	18.5
MELM [53]	val1	val2	13.4
P-MIDN+MGSC	val1	val2	16.3
P-MIDN+MGSC	train1k+val1	val2	29.0
OM+MIL+FT [61]	val1	val2	10.8
WCCN+FRCNN [16]	-	-	16.3
SP [56]	train1k+val1	val2	12.1
PCL-Ens.+FRCNN [43]	train1k+val1	val2	19.6
CS+AS+MIL+Seg+FT [58]	val1	val2	12.1
P-MIDN+MGSC+FRCNN	val1	val2	20.7
P-MIDN+MGSC+FRCNN	train1k+val1	val2	31.9

TABLE V

COMPARISON OF AP PERFORMANCE (%) ON MS-COCO

Method	Training data	Testing data	AP ₅₀	AP
ML-LocNet [62]	train2014	val2014	16.2	-
PCL [43]	train2014	val2014	19.4	8.5
MELM [53]	train2014	val2014	18.8	7.8
CSC [35]	train2017	minival	20.3	10.2
WS-JDS [17]	train2017	minival	20.3	10.5
C-MIDN [10]	train2014	val2014	21.4	9.6
PG-PS [38]	trainval135k	minival	20.7	-
P-MIDN+MGSC	train2014	val2014	27.4	13.1
P-MIDN+MGSC	train2014	minival	27.7	13.4
P-MIDN+MGSC	train2014	test-dev	28.0	13.5
WCCN+FRCNN [16]	train2014	test-dev	12.3	-
PCL-Ens.+FRCNN [43]	train2014	val2014	19.6	9.2
MEFF [41]	train2014	val2014	19.3	8.9
ODGA+SSD [59]	train2014	test-dev	13.6	-
WSOD ² [30]	train2014	val2014	22.7	10.8
P-MIDN+MGSC+FRCNN	train2014	val2014	28.5	13.2
P-MIDN+MGSC+FRCNN	train2014	minival	29.0	13.5
P-MIDN+MGSC+FRCNN	train2014	test-dev	29.2	13.5

either the train2017 or the trainval135k splits (around 118K images).

D. Ablation Study

1) *Influence of Different Components:* We analyze the influence of different components on the PASCAL VOC 2007 test set as indicated in Tab. VI. Our baseline model [7] (denoted by M1) is constructed by a basic MIDN followed by an online refinement module. By introducing segmentation IOUs (M2), the mAP increases to 51.3%. This demonstrates

TABLE VI

ABLATION STUDY ON PASCAL VOC 2007 Test SET

Method	Seg.	Multiple	Components Coupled	Pyramidal	MGSC	mAP
M1	-	-	-	-	-	48.5
M2	✓	-	-	-	-	51.3
M3	✓	✓	-	-	-	51.5
M4	✓	✓	✓	-	-	51.9
M5	✓	✓	-	✓	-	52.9
M6	✓	✓	-	✓	✓	53.9

the importance of segmentation priors for selecting pseudo GTs. Furthermore, when multiple ($K = 3$) *independent* MIDNs are employed (M3), the improvement is marginal. We also train a model (M4) using the proposal removal method of coupled MIDN [10] with the same segmentation masks used in our method. Compared with M3, M4 improves the mAP performance by 0.4%. When the proposed pyramidal structure (M5) is used, the performance improves significantly. Our P-MIDN outperforms M4 by 1.0% in mAP. Note that for M3, M4 and M5, we select the highest scoring proposal from multiple MIDNs as the pseudo GTs. With the help of our proposed mask guided self-correction (MGSC) method, the mAP of our proposed approach (M6) reaches 53.9%.

Fig. 6 demonstrates the per-class performance of models with coupled MIDN, pyramidal MIDN and pyramidal MIDN with MGSC. The model with pyramidal MIDN outperforms that with coupled MIDN on 15 out of 20 categories. When MGSC is utilized, the performance on some animal categories significantly increases, *e.g.*, bird, cat and dog.

2) *Influence of Segmentation Masks:* Since segmentation masks provide important prior information for our proposed method, we conduct several experiments to analyze their impact on the detection performance. The IRN [24] can generate both the semantic and instance segmentation masks for the training images. Tab. VII shows a comparison of our proposed P-MIDN+MGSC with different segmentation types. The instance segmentation masks could provide stronger priors in some cases. For instance, when objects of the same class are adjacent to each other in the same image, it is hard to separate them with only semantic segmentation masks. High-quality instance segmentation masks could help alleviate this issue for our method. We can see that with the instance segmentation masks, our method further improves the performance by 1% in mAP. Note that in other experiments we only use semantic segmentation masks to conduct a fair comparison with other methods which also use semantic segmentation masks.

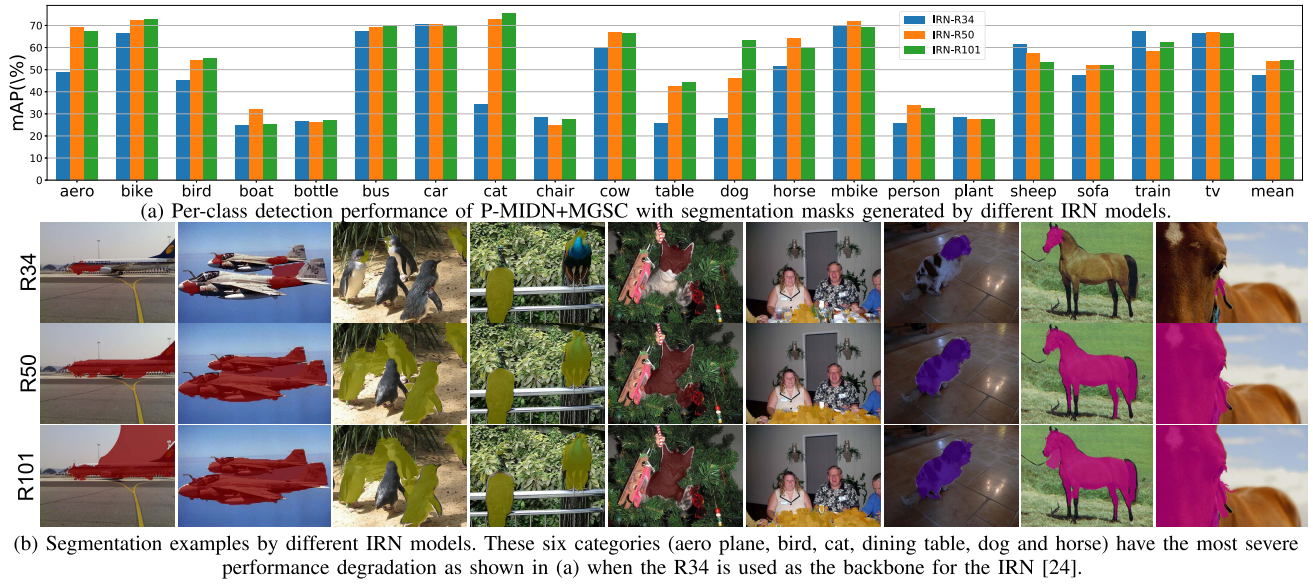


Fig. 7. Influence of segmentation masks on detection performance.

TABLE VII
RESULTS OF OUR PROPOSED METHOD USING SEMANTIC AND
INSTANCE SEGMENTATION MASKS FROM IRN

Mask type	mAP
Semantic mask	53.9
Instance mask	54.9

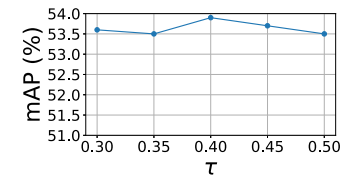
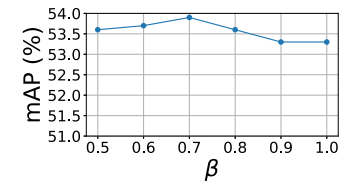
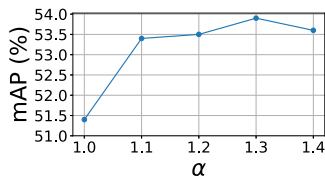
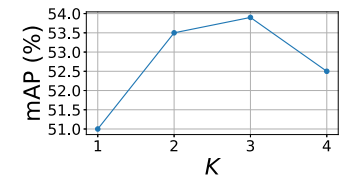
Fig. 9. Influence of different fitness upper bound τ .

TABLE VIII
COMPARISON OF DIFFERENT METHODS UTILIZING IRN'S OUTPUTS

Method	mAP
IRN+FRCNN	47.2
P-MIDN+MGSC	53.9
P-MIDN+MGSC+FRCNN	55.0

Fig. 10. Influence of different IOU threshold β .Fig. 8. Influence of different scaling factor α .Fig. 11. Influence of different number of MIDNs K .

In order to validate the improvement is mainly from our proposed method rather than the segmentation masks from the IRN, we train a Fast R-CNN by using the fitted bounding-boxes from the instance segmentation masks produced by the IRN as pseudo ground-truths (denoted by IRN+FRCNN). As shown in Table VIII, our proposed P-MIDN+MSGC outperforms the baseline IRN+FRCNN by 6.7% in mAP. This is because the pseudo ground-truths generated from the instance segmentation masks of the IRN contain noise. Training a detector with such noisy pseudo ground-truths

leads to unsatisfactory detection results, therefore demonstrating the necessity of our proposed P-MIDN+MGSC. By re-training a FRCNN with the pseudo ground-truths generated by P-MIDN+MGSC, the performance boosts to 55.0%.

We also conduct an experiment to analyze the impact of the outputs of the IRN on our proposed method. We use different network backbones (*i.e.* ResNet-34 (R34), ResNet-50 (R50) and ResNet-101 (R101) [63]) to train IRN. Fig. 7(a) shows the detection performance of our proposed method using segmentation masks generated by different IRN models.

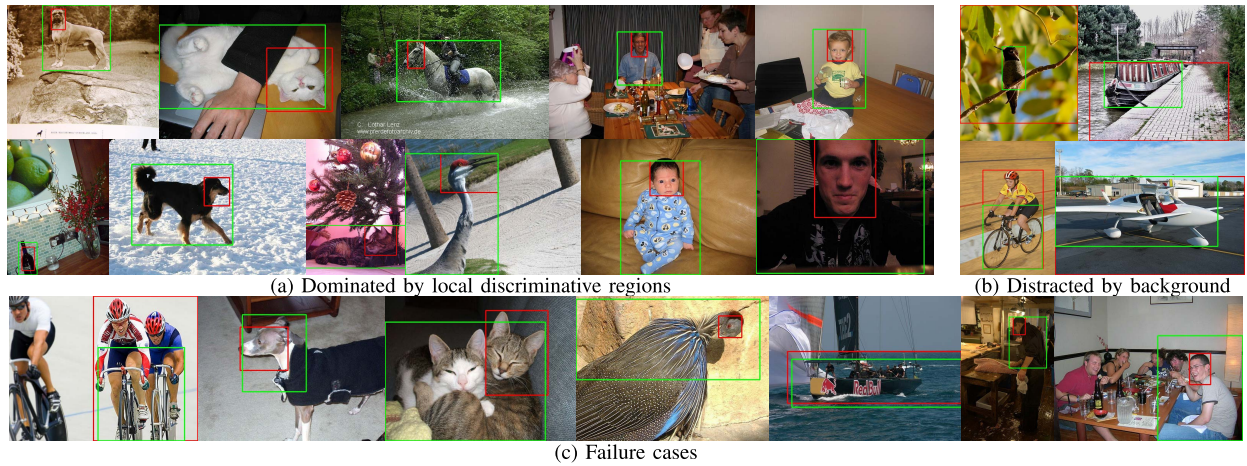


Fig. 12. Visualization of detection results on the PASCAL VOC 2007 Test set. Red bounding boxes indicate the top-scoring detections of our baseline, while the green ones are produced by our proposed P-MIDN.

The IRN trained with a better backbone generally produces segmentation masks of higher quality, providing stronger prior for our proposed method. Replacing the R50 backbone (default choice in IRN [24]) with R101, the mAP performance of our proposed P-MIDN+MGSC improves from 53.9% to 54.3%. When R34 is used as the backbone, the quality of segmentation masks decreases, resulting in performance degradation of our proposed method. For some categories like aero plane, cat and dog, the detection performance decreases drastically due to poor segmentation masks generated by the IRN with backbone R34. Fig. 7(b) shows segmentation examples of top 6 object categories with the most severe performance degradation when the R34 is used as the backbone in IRN.

3) *Influence of Scaling Factor*: In our proposed P-MIDN, an adaptive threshold is utilized to remove some proposals located around local discriminative regions. The scaling factor α controls the growth speed of the adaptive threshold (see Sec. III-B). Fig. 8 shows the mAP results of the proposed method with different α . It can be seen that our proposed method is robust when α ranges between 1.1 and 1.4. The performance of the proposed method reaches the peak at $\alpha = 1.3$. For $\alpha = 1.0$, the proposals around local discriminative regions cannot be effectively removed, resulting in a large performance degradation.

4) *Influence of Fitness Upper Bound*: The fitness upper bound τ is another hyper-parameter for P-MIDN training. It determines the boundary for high-quality proposals and low-quality proposals (see Sec. III-B and Sec. III-C). Fig. 9 shows the results of our proposed method with different τ . We achieve the best mAP when τ is set to 0.4. Due to the imperfect segmentation masks, connected components may cover object parts, multiple objects or background. A good trade-off is achieved at $\tau = 0.4$ for these situations.

5) *Influence of Local Refinement IOU Threshold*: The IOU threshold β controls the range of local refinement. Higher β indicates fewer proposal candidates are taken into consideration for local refinement. When β is set to 1, the local refinement is not implemented. As shown in Fig. 10, the best detection results are obtained when β is set to 0.7.

As β becomes smaller, the performance decreases marginally. We believe β would be even more robust when better segmentation masks are provided.

6) *Influence of the Branch Number of P-MIDN*: The total branch number K determines how many MIDNs are adopted in the P-MIDN. We use different K (i.e. $K = 1, 2, 3$ and 4) while fixing all the other hyper-parameters in our method. When K set to 1, the basic MIDN is used rather than the proposed P-MIDN. As shown in Fig. 11, we only obtain an mAP of 51.0%. The best performance is attained when 3 MIDNs are used to construct the P-MIDN. When K increases to 4, our proposed method has some performance degradation. This is because the 4-th MIDN could be trained with over-loose proposals when K increases from 3 to 4.

E. Inference Speed

During inference, only the network backbone and online refinement module are required. Thus, our proposed method is as efficient as any other OICR-based methods [7]–[11], [17], [18]. Specifically, our proposed P-MIDN+MGSC and P-MIDN+MGSC+FRCNN take about 1.4 and 1.2 seconds to process one image, respectively. Both models are evaluated on the PASCAL VOC 2007 test set with 5 multi scales and horizontal flip using a single NVIDIA Tesla V100 GPU.

F. Qualitative Results

Fig. 12 shows some qualitative examples of our baseline model [7] and our proposed method. It can be clearly seen that the baseline model is prone to detect local discriminative object regions, e.g. animal heads. Benefiting from the proposed P-MIDN, we can better localize whole objects (as shown in Fig. 12(a)). For some categories, e.g. aeroplane and bicycle, the baseline model may produce overly-loose bounding boxes which include background regions, while our proposed approach provides tighter detection results (as shown in Fig. 12(b)). This indicates that our mask guided self-correction method effectively excludes overly-loose pseudo GTs. Some failure cases are provided

in Fig. 12(c). When objects with the same class are closely adjacent to each other, our method might fail to detect different instances. In addition, occlusions and drastic appearance variations (e.g. dogs wearing clothes) still remain challenging for WSOD.

V. CONCLUSION

In this paper, we first proposed a novel pyramidal multi-instance detection network, namely P-MIDN, for weakly supervised object detection with only image-level annotations. Each MIDN in the proposed P-MIDN performs proposal removal to reduce the exposure of its subsequent MIDN to local discriminative regions during training. Moreover, a mask guided self-correction (MGSC) method is further designed to modify pseudo ground-truth bounding boxes provided by the P-MIDN via our proposed self-labeling or local refinement mechanisms. Benefiting from the MGSC, we obtain higher-quality pseudo ground-truths and use them to train our detection network. Extensive experiments on five commonly used benchmarks validate the superiority of our proposed method.

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